Predicting Credit-Card Default: A Comparative Study of Machine-Learning Classifiers

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Abstract

A missed defaulter costs an issuer far more than a false alarm, so a default model must be judged at an operating point, not by ranking alone. We predict next-month default on the UCI *default of credit card clients* data: 30,000 Taiwanese card holders, 29,965 after cleaning, 22.1% defaulters. The study compares KNN, regularised logistic regression, discriminant and naive Bayes classifiers, pruned trees, random forests, boosted ensembles, support-vector machines and a neural network on one fixed 70/30 stratified split, with preprocessing and resampling inside cross-validated pipelines. Preliminary runs price the threshold above the model: moving the logistic cut-off from 0.5 to the 0.15 implied by an assumed 5:1 cost cuts expected cost from 7,063 to 5,126, while the best model (XGBoost, test AUC 0.789 against 0.764) saves a further 145.

Contribution statement. Group B19. Sections 1–4 and 6, the exploratory analysis, the shared split-and-evaluation contract (`code/common.py`), the aggregation and the threshold-and-cost analysis were done jointly, one third each. Each member owns three to four model families in Section 5 and runs each one end to end — data, tuning, interpretation, presentation: RUAN Hangjing the KNN benchmark, the logistic family and the PCA/K-means segmentation; LU Xinyu the pruned tree, random forest and bagging, gradient boosting and XGBoost; LI Jingwen the support-vector machines, the neural network, LDA/QDA/naive Bayes and the class-imbalance study with its leakage demonstration. The owner of every result file is recorded in `results/`. Not shared with any other module.

1 Introduction and problem statement

Unsecured consumer credit is one of the largest sources of credit losses for retail banks: when a card holder misses the minimum payment, the issuer loses the outstanding balance and incurs collection costs and a higher regulatory-capital charge. Predicting *which* customers will default next month lets a bank act early — tighten limits, reprice, prioritise collections or decline an application — while leaving reliable customers untouched. The task is binary classification with two very different error costs: a missed defaulter (false negative) costs several times more than a false alarm (false positive), so the operating threshold, and not only the ranking quality, matters.

This project builds and compares classifiers that predict next-month default from a customer's credit limit, demographics and six months of billing and repayment history. Rather than searching for "the best" model, the objective is a **comparative study**: how much a simple, interpretable model already captures, how much non-linear and ensemble methods add, how class imbalance should be handled, and what the results mean for a risk manager.

MH6805 Machine Learning in Finance, Nanyang Technological University. Group project proposal, Group B19.

2 Dataset

Source and target. *Default of credit card clients* [1], UCI Machine Learning Repository, dataset 350 (`https://archive.ics.uci.edu/dataset/350/default+of+credit+card+clients`): real records of 30,000 card holders of a Taiwanese bank, April to September 2005, in 25 columns (an ID, 23 features, the target). The raw file is public, so it is not attached; the cleaning and feature-engineering code that reproduces the processed data ships with the final report. The target `DEFAULT` is 1 if the client defaulted on the October 2005 payment: **6,636 defaulters (22.1%) against 23,364 non-defaulters (77.9%)** — a moderately imbalanced problem in which a trivial "never default" rule is already 77.9% accurate.

Variables. The 23 raw predictors are the credit limit `LIMIT_BAL` (NT$); the demographics `SEX`, `EDUCATION`, `MARRIAGE`, `AGE`; six repayment-status codes `PAY_1`–`PAY_6` (September back to April; -1 paid duly, $k \geq 1$ a delay of k months, observed range $-2 \ldots 8$); and the monthly bill and payment amounts `BILL_AMT1`–`BILL_AMT6`, `PAY_AMT1`–`PAY_AMT6`. From these we engineer six features encoding credit-risk domain knowledge, giving 29 predictors in all: utilisation (`UTIL_1`, `UTIL_MEAN`), the average payment-to-bill ratio (`PAY_RATIO_MEAN`), months in arrears (`N_MONTHS_DELAYED`), the longest delay (`MAX_DELAY`) and the six-month balance trend relative to the limit (`BILL_TREND`).

Data quality. There are no missing values. We dropped 35 exact duplicate rows (29,965 clients remain), recoded the undocumented category codes (`EDUCATION` $\in \{0, 5, 6\}$, `MARRIAGE` $= 0$) to "others", and kept the undocumented repayment codes -2 and 0: they mean "no consumption" and "revolving credit" and carry information, their default rates being 13.7% and 12.9%. Bill amounts can be negative (6.4% of clients) and 17.2% paid nothing in the latest month; both are genuine and retained. The six bill amounts are collinear ($r \approx 0.8$–0.95), so the linear models are regularised and their coefficients read with care.

Why no external data. Enriching the panel with an exogenous series was considered and rejected on mechanical grounds: a join needs a key and this file has none — no geography, no industry, no client-level date — and all 30,000 clients share one April–September 2005 window, so any macroeconomic series would enter as a single constant column and add nothing. The room for enrichment is therefore feature engineering on the existing panel, which is what the six derived features do.

Exploratory findings. All statistics below use the training partition only ($n = 20{,}975$); the full set of exploratory plots is in the code repository. The most recent repayment status is by far the strongest single signal: the default rate rises from 12.9% for clients on revolving credit in September to 33.9% after one month of delay and 67.6% after two. It falls monotonically with the credit limit (32.1% in the lowest quintile against 13.7% in the highest), is mildly higher for men and for lower education, and is U-shaped in age. The months in arrears and the longest delay are the most informative engineered features (univariate AUC 0.72 and 0.71); utilisation and the payment-to-bill ratio carry only weak signal (0.55–0.59), and linear correlations with the target are weak ($|r| \leq 0.40$) — which suggests non-linear methods may add value.

3 Objectives

1. **Quantify the value of model complexity.** Against a KNN benchmark and a logistic baseline, measure how much each family in Table 1 improves threshold-free ranking quality (test ROC-AUC, average precision), and whether the gaps are statistically meaningful (bootstrap confidence intervals for AUC differences).
2. **Choose the operating point for the business, not the default 0.5.** Report each model's confusion matrix at 0.5 and at thresholds tuned on the training data to maximise F1 and to minimise an assumed 5:1 cost of false negatives to false positives.
3. **Handle class imbalance correctly and price its effect.** Compare no resampling, class weighting, SMOTE and under-sampling applied *inside* the folds, and demonstrate the inflated results obtained when resampling wrongly precedes the split.
4. **Interpret, not only predict.** Identify the strongest predictors (odds ratios, permutation and impurity importance), check them against credit-risk intuition, and profile customer segments with PCA and K-means.

Table 1: Proposed methods, rationale and tuning grids.

Method (lecture)	Why it is included	Tuning
K-nearest neighbours (B4)	Course benchmark; non-parametric, scale-sensitive	$K \in \{5, \ldots, 151\}$, uniform vs distance weights
Logistic regression: plain, ridge, lasso (B5, B6)	Interpretable baseline; coefficients as odds ratios, lasso selects	penalty strength C on a log grid
LDA, QDA, naive Bayes (B6)	Generative rivals; price their distributional assumptions	QDA regularisation, NB smoothing
Decision tree, cost-complexity pruning (B7)	Transparent rules; exhibits over-fitting	`ccp_alpha` path, max depth
Random forest and bagging; gradient boosting and XGBoost (B7)	Variance reduction and de-correlation, then bias reduction; permutation importance; usually the strongest tabular learners	`max_features`, `min_samples_leaf`, 500 trees; depth, learning rate, trees, subsampling
Support-vector classifier, RBF-kernel SVM (B8)	Margin-based boundary; hinge against logistic loss	C, γ on a stratified subsample, refitted on all training rows
Feed-forward neural network (B9)	Tests whether depth helps on tabular data	layers, L2 penalty, learning rate, early stopping
Class-imbalance handling (B4)	Weights, SMOTE, under-sampling inside the folds; leakage demo	sampling ratio, SMOTE k
PCA, K-means, hierarchical clustering (B2, B3)	Segmentation; components and labels as extra features	components, K by elbow and silhouette
TreeSHAP (beyond the syllabus)	Importance says which variable matters; Shapley values say in which direction it pushes each prediction	`shap` TreeExplainer on the tuned XGBoost; beeswarm and dependence plots
Monotonically constrained XGBoost (beyond the syllabus)	More delay can never lower predicted risk, so the model reads like a scorecard; prices that constraint in AUC	`monotone_constraints` ± 1 on delay, utilisation, limit, payment ratio

4 Proposed methods

Protocol. The cleaned data are split once, stratified by the target, into a 70% training set (20,975 clients) and a 30% test set (8,990 clients) under seed 2607526; the test set is touched exactly once per model, for the final evaluation. All preprocessing — standardisation, one-hot encoding of `SEX`/`EDUCATION`/`MARRIAGE`, any resampling, the PCA and K-means features — sits inside a scikit-learn `Pipeline`, so it is re-fitted within each fold of the 5-fold stratified cross-validation that selects hyperparameters and no information from validation or test rows leaks into training. The tuning criterion is cross-validated ROC-AUC; average precision, Brier score and per-class precision/recall/F1 are reported alongside, with all models' ROC and precision–recall curves overlaid on one figure, bootstrap intervals for the AUC differences and calibration plots. Method definitions follow the course text [2].

Why a 5:1 cost ratio. The ratio is an assumption, anchored three ways. It is the cost matrix of the UCI Statlog German Credit benchmark [3] (a bad customer accepted costs 5, the reverse 1). A back-of-the-envelope on our training data implies more: a defaulter's median outstanding bill is NT$19,868, so a missed default at a 50–70% loss-given-default costs NT$10–14k, against roughly NT$340 of forgone monthly margin on a good customer — 10:1 or more, making 5:1 conservative. A sensitivity analysis from 1:1 to 20:1 is also reported: the same three ensembles lead from 3:1 to 20:1, though their order swaps, while the optimal threshold moves from 0.27 to 0.05. The lecture's closed form for the cut-off implied by a cost matrix, $t = C_{FP}/(C_{FN} + C_{FP})$, gives $1/6 \approx 0.167$ at 5:1, and our empirical search on training out-of-fold probabilities returns 0.15 — reproducing the theoretical value. That formula holds only for calibrated probabilities, which is why calibration is reported too.

Table 2: First-round results. CV AUC is the mean 5-fold out-of-fold AUC on the training set; all other columns are on the held-out test set (8,990 clients). Thresholds are tuned on training out-of-fold probabilities; cost is the test expected cost at a 5:1 false-negative : false-positive ratio, against 7,001 for flagging everybody and 9,945 for flagging nobody.

Model	CV AUC	Test AUC	Test AP	F1 thresh.	Recall / FPR at cost thresh.	Cost at 0.5	Cost at cost thresh.
Logistic regression (baseline)	0.760	0.764	0.523	0.30	0.74 / 0.36	7,063	5,126
K-nearest neighbours ($K = 101$)	0.770	0.771	0.542	0.25	0.75 / 0.39	6,854	5,219
Pruned decision tree	0.769	0.776	0.535	0.28	0.76 / 0.38	6,933	5,085
Random forest (500 trees)	0.785	0.787	0.576	0.31	0.76 / 0.37	6,598	4,998
XGBoost	0.786	0.789	0.578	0.27	0.78 / 0.40	6,642	**4,981**
Feed-forward neural network	0.776	0.773	0.551	0.27	0.71 / 0.32	6,625	5,136
SVM, RBF kernel	0.761	0.762	0.494	0.42	0.69 / 0.29	9,316	5,155

5 Preliminary results

The pipeline, the shared evaluation function and a first round of every model family have been implemented and run on the fixed split; Table 2 reports the seven main models. A plain logistic regression on all 29 features reaches a **test AUC of 0.764** (average precision 0.523 against a prevalence of 0.221). At threshold 0.5 it is 81.5% accurate (majority-class accuracy 77.9%) but recovers only **32% of the defaulters** and costs 7,063 — more than the 7,001 of flagging every customer; the cost-minimising threshold 0.15 recovers 74% at a 36% false-positive rate, for 5,126. Tree ensembles lift the ranking to AUC 0.787–0.789 and average precision 0.58. Against the KNN benchmark (0.771) only the tree-based models and, marginally, the neural network are better; the linear models and the RBF-kernel SVM fall below it, and explaining those shortfalls (limited non-linearity; an SVM tuned on a 6,000-row subsample, Platt-scaled) is part of the planned interpretation. Three working hypotheses follow: the linear model already captures most of the signal, non-linearity adding only 0.025 AUC (paired bootstrap 95% CI [0.018, 0.031]); the threshold dominates the choice of model; and the ranking should be explainable by the interaction of repayment delays with limit and utilisation, which feature-importance and partial-dependence analysis will test.

6 Work plan and division of labour

14–18 Sep: cleaning, feature engineering, EDA, the shared evaluation code and this proposal. **19 Sep – 10 Oct:** the model families, each owner end to end — KNN, the logistic family, PCA and clustering (RUAN Hangjing); tree, random forest and bagging, gradient boosting, XGBoost (LU Xinyu); SVM, neural network, LDA/QDA/naive Bayes, class-imbalance study (LI Jingwen). **11–20 Oct:** aggregation — comparison table, overlaid ROC/PR curves, bootstrap confidence intervals, threshold and cost analysis, calibration. **21–26 Oct:** drafting the final report in this template (≤ 7 pages including references) and the slides, which are due noon on the day before the presentation. **Presentation week (27 Oct, to be confirmed):** each member presents the models they own and all answer questions; the report, code and data follow two days later. Every stage but the second is joint.

References

[1] Yeh, I.-C. & Lien, C.-H. (2009) The comparisons of data mining techniques for the predictive accuracy of probability of default of credit card clients. *Expert Systems with Applications* **36**(2):2473–2480.

[2] James, G., Witten, D., Hastie, T. & Tibshirani, R. (2021) *An Introduction to Statistical Learning*, 2nd ed. New York: Springer.

[3] Hofmann, H. (1994) *Statlog (German Credit Data)*. UCI Machine Learning Repository, dataset 144.